SIMULATION · NOT A FIELD SYSTEM

AFTERSHOCK

A two-robot search-and-rescue agent that maps a collapsed building, predicts what may fail next, and sends help only along routes it has checked.

Every move traces to a reading; a scene change revokes the route; **physics and a human, not a language model, authorise movement.**

PROBLEM

After an earthquake, rescuers cannot see inside a collapsed building, and an aftershock can bring down more of it on the people who go in. Sending a person or a heavy machine in blind wastes the first hours and adds risk.

Time is the constraint: "the vast majority of survivors are brought out within 24 hours by local teams", and more than 90% within three days (Ilan Kelman, UCL disaster researcher, to AFP, 9 Feb 2023[1]). A robot that maps safely in those first hours saves the time a blind entry loses.

USERS

- **USAR team lead / operator** (primary): decides where it is safe to send people or machines; approves risky moves. Often hands busy, so every command works by voice or typed.
- **Trapped casualty:** no screen and no login. Hears the Scout call out, answers by tapping or speech, gets supplies and spoken guidance in their language.
- **Next team in:** reads the handover report: what is safe, who is where, what was approved.

WHAT IT DOES

- **Scout** explores and maps: cells are *seen*, *inferred* or *unknown*; only seen-free cells can carry a route.
- Finds casualties from pose, thermal and sound (no faces); predicts *stable / at-risk / collapsed* blocks.
- **Responder** delivers water, a first aid kit or a radio, clears light debris with a 0.5 kg-payload arm.
- Voice: operator commands, casualty translation, cited first-aid steps. Handover report at the end.

HOW IT WORKS

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MuJoCo sensors (10 Hz) → perception map (2 Hz) → local A* route → MuJoCo physics check → move permit → human gate if risky → Responder action → log/decision
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Perceive. Scout ray-cast LiDAR (360 rays), depth/RGB, a 16×12 thermal grid and acoustic bearing build an occupancy map and a fused casualty estimate with a confidence.

Check. MuJoCo simulates an aftershock against the route; a route through a block that would move is rejected and re-planned. Passing earns a **move permit**.

Decide. Local A* proposes a route over seen cells, keeping 0.15 m from walls. A confidence gate picks *act*, *re-look* (send the Scout back) or *refer to the operator*; it never guesses silently.

Adapt. Any sensed change (a block lands, debris moves) **revokes** the permit; the Responder stops, the Scout re-scans, and a new route is verified and explained in one line.

CRITICAL FAILURE, HANDLED ON SCREEN

Stale map: debris moves after a route was approved. The LiDAR scan diff revokes the permit, the Responder stops on the spot, and travel resumes only after every route cell is freshly re-scanned and physics-checked again.

Measured (B1 run, seed 3): block lands at 50.1 s → permit revoked 50.1 s → re-granted after a fresh check 52.3 s → Responder moving on a new route 53.1 s (sim time).

WHERE EACH MODEL RUNS (HARDWARE AND LATENCY)

Function	Model / method	Runs on	Measured latency / budget
Routing, confidence gate, handoff	Local A*, rule-based gate	**Device**	under 50 ms per step
Offline planner (goal reading)	Gemma 4 E2B (gemma4:e2b, Q4_K_M) via Ollama 0.40.15	**Device**	1.8–2.8 s warm, 19 s cold (pre-warmed at start); 4.1 GB resident. Budget: 5,000 MB, 4,000 ms per step, shown live
Physics: world, sensors, route check	MuJoCo 4.0.0	**Device**	about 0.6–0.9 ms per 2 ms physics step (2–4× real time, 3 scenes)
Mapping, casualty fusion, hazards	Python (grid, scan diff, keyframes)	**Device**	15.7 ms per sensor frame (970-frame replay)
Camera check, casualty talk, translation, report	gemini-3.8-flash	**Cloud, optional**	Advisory only: never grants movement. Each call logs a receipt with its wall time; capped at 300 calls/h. With `GEMINI_MODE=off` every feature falls back to a local stand-in and the full run still completes.
Spatial look points, approval routing	gemini-robotics-er-2-preview, gemini-3.5-flash-lite	**Cloud, optional**	
Operator voice, scene from photos	gemini-3.8-live + flash-tts, gemini-4-argon	**Cloud, optional**	

Device = the demo laptop: CPU only, 8 GB RAM, no GPU (WSL2, 12 threads). Target robot compute: Jetson Orin NX 16 GB, so the 4.1 GB planner fits beside mapping.

STACK

- **Sim:** MuJoCo 4.0.0, Python 3.12; scenes: backup, building, tower, multi-level (L1–B3).
- **Lanes:** sim, perception, planner, voice: async Python on one validated topic bus (FastAPI + uvicorn, WebSocket /ws); optional sim-in-own-process mode.
- **Console:** TypeScript 5.9, Three.js 0.181, Vite 7; browser speech for voice; Playwright tests.
- **Local LLM:** Ollama + Gemma 4 E2B. Cloud: Google GenAI SDK.

STATES AND INTERFACES

- **Map cell:** seen / inferred / unknown; **hazard:** stable / at-risk / collapsed.
- **Responder:** idle, moving, arrived, blocked, frozen (shaking).
- **Gate:** act, re-look, refer. **Permit:** granted, revoked.
- Lanes talk only through typed bus topics (e.g. `route/proposed` → `route/verified` → `permit/route`); the browser may write only `operator/command` and `gate/answer`.

SAFETY RULES AND LIMITS

- Risky moves (near a casualty, into an at-risk cell) and any emergency call need explicit operator approval.
- A model answer never authorises movement; physics and the permit do.
- No faces, no diagnosis: casualty vitals are simulated and labelled; first aid is cited guidance for a human.
- Every decision is logged with its perception, model, physics result and outcome.
- **Limits:** simulation only; hazard and casualty outputs are estimates, not engineering or medical advice; sim robots are scaled down (a real Jackal needs ~0.8 m doorways); real use needs independent validation.

EVIDENCE: WHAT WE MEASURED

- **End-to-end runner** (one command, auto-approve, local models only): B1 seeds 3, 5, 18 and 19 pass **17/17** checks.
- **Multi-level descent** to B2: seeds 0–4 pass **9/9**: the Scout descends, maps a dropped slab, stops, re-routes, reaches the casualty.
- **Test suites:** about 900 Python tests and 36 browser tests on the main branch.
- **Failures we found and fixed:** the most common miss was "supply placed within 0.5 m of the casualty" (the collision ray read the casualty as a wall; re-plans started from a stale cell). After both fixes, seeds 3 and 5 pass **17/17**.

HARDWARE CONCEPT (NOT BUILT)

Scout: AgileX LIMO Pro, Jetson Orin Nano, 2D LiDAR (12 m), depth, FLIR Lepton 3.5, 4-mic array, IMU; ~2.3 h.

Responder: Clearpath Jackal, Jetson Orin NX 16 GB, Livox Mid-360 3D LiDAR, OAK-D Lite, Lepton 3.5, mics + speaker, Kinova Gen3 lite arm; ~26 kg, ~2 h.

TRACK FIT

Perception-driven action (25%): each log line names the reading behind it; a re-look is a decision, not a fallback.

Adaptation (25%): an aftershock or a block dropped live from the console revokes the route; the new one is drawn with its reason.

Autonomy (15%): core decisions run on the device within a stated, metered budget; cloud models are advisory.

Impact and quality (35%): the stale-map failure is handled on screen; the video is one take with one unscripted change.

[1] AFP via Daily Sabah, 9 Feb 2023: dailysabah.com/turkey/chances-of-survival-may-be-up-to-a-week-after-earthquakes-experts/news (expert comment, not a study). Sources: docs/brief.md, docs/aftershock.md, docs/hardware.md, planner/budget.json, coord/CONTRACT.md, coord/gem.py; runs and benchmarks from the team board, 8 Nov 2026.